Fiscal Year:	FY 2006	**Task Last Updated:**	FY 11/28/2007
PI Name:	Diddams, Scott A. Ph.D.		
Project Title:	Compact Femtosecond-Laser-Based Optical Synthesizers for Precision Measurements and Atomic Clocks in the Optical Domain		
Division Name:	Physical Sciences		
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Flight Program:			
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Atomic frequency standards that are being designed to fly aboard the International Space Station have projected fractional frequency uncertainties below 1e-16 with estimated instability as low as 3e-15 at 1 s of averaging. At the same time, emerging optical frequency standards based on laser-cooled atoms and ions promise even superior fractional frequency stability (<2e-16 at 1s of averaging) and uncertainty (1e-18). While it is valuable to operate these microwave and optical standards in isolated experiments, powerful tests of fundamental physical forces and symmetries can be designed around the comparison of standards operating at extremely different frequencies (i.e. microwave versus optical frequencies). Furthermore, there are several space-borne scientific and technical applications that could benefit from the extremely low fractional frequency instability provided by optical frequency standards if it could be transferred to the microwave domain. However, until very recently an appropriate high-performance synthesizer did not exist for

	phase-coherently comparing and linking frequency standards in the optical and microwave domains. This research will attempt to fill this gap with science-enabling technological developments for high performance laser-cooled atomic time and frequency standards. The central objective is to develop femtosecond-laser-based technology into a compact, versatile, and robust optical frequency synthesizer. This ground-based study will involve the construction and testing of compact, low-power femtosecond laser sources, the stabilization of these lasers, and the determination of their relevant properties and fundamental limits. The outcome of the proposed research is anticipated to be a general-purpose device that can synthesize microwave frequencies from optical frequencies (and vice-versa) without degradation of the stability and accuracy of the reference input. This should prove valuable for fundamental physics experiments associated with the comparison of atomic clocks based on different transitions as well as the generation of very stable microwave signals.
Task Description:	The central objective of this research is to develop femtosecond-laser-based technology into a compact and versatile optical frequency synthesizer. This ground-based study will center around the construction, testing, and optimization of a new femtosecond laser source that directly emits an octave-spanning spectrum. This will improve the robustness of these devices, making it especially suitable for applications in space. The outcome of the proposed research will be a general-purpose device that can synthesize microwave frequencies from optical frequencies (and vice-versa) without degradation of the stability and accuracy of the reference input. The femtosecond-laser-based optical frequency synthesizer is projected to support stabilities of 1e-16 at 1 s, and have a frequency uncertainty approaching 1e-18. This should prove valuable for fundamental physics experiments on earth and in space in at least three major areas. Because they hold potential for significant stability and accuracy gains, there is good reason to believe that atomic clocks based on optical transitions will be the next generation of primary atomic frequency standards. The femtosecond-laser-based synthesizer proposed here is a crucial part of these future clocks. The potential for wide-spread positive impact can be extrapolated from the many ways in which precision time and frequency standards already play a fundamental role in the personal, commercial and scientific aspects of our society. Many civilian applications, and all branches of the military make extensive use of high precision atomic clocks. Applications of precision frequency standards range from the obvious¯time keeping and references for navigation¯to advanced future uses such as space-based atomic clocks for worldwide time/frequency distribution and navigation. High stability frequency references play critical roles in communication and navigation systems including, anti-jamming, friend or foe identification, spread spectrum communications, and increased security. These new frequency references coupled with the femtosecond-laser-based optical synthesizer will allow us to open the door to unexplored regions of science and space-time. Examples include searches for new physics such as graininess of space, and any time variation of atomic structure and fundamental constants. Some recent astronomical observations show evidence for a time variation of the fine structure constant, which if verified by earth or space-based measurements would have profound implications for our understanding of nature.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Over the past 50 years (since their inception) atomic clocks have had a tremendous impact on the science, technology and society of our country. This project is part of the development of the next generation of atomic clocks, which are based on optical frequencies. As such, it is fully anticipated that this research will impact the fields of communications, navigation and basic science. Atomic clocks have been critical sources for advanced navigation and communications systems. This work will support the development of these technologies.
Task Progress:	We have continued to make significant and positive progress on this project in FY05. Planned milestones have been achieved and exceeded in some cases. Specific summaries in areas of progress are given below. 1. A significant advance was made during the past year in the generation of extremely low noise (less than or equal to -110 dBc/Hz @1 Hz offset from 10 GHz carrier) microwave signals. Such low noise signals were generated from an optical references oscillator (laser stabilized to a high-finesse cavity) that had its optical frequency divided down to 10 GHz using the octave spanning femtosecond-laser-based optical synthesizer. This level of noise close to the carrier is ~40 dB lower than can typically be achieved using conventional microwave oscillators and frequency sythesis techniques. In the time domain, the corresponding instability of the 10 GHz signal was ~3e-15 at 1 s of averaging (limited by the optical reference oscillator), while the residual noise of the femtosecond laser synthesizer on this same time scale was measured to be <1e-16. These levels of perfomance were verified by comparing multiple optical references and femtosecond laser synthesizers. 2. On another front, we have had good success in generating more efficient and compact femtosecond laser optical synthesizers based on Cr-doped forsterite gain media. The Cr:forsterite laser operates at 1.3 microns and is compatible with low-loss optical fiber trasnmission. This femtosecond laser also has the benefit of being pumped by an efficient and relatively compact fiber laser, making it a candidate for a transportable frequency comb (compared to Ti:sapphire which requires a much more expensive, heavy and less efficient pump source). We have also worked with a US industrial partner (OFS Labs) on the implementation of novel nonlinear optical fibers that allow us for the first time to tailor the nonlinearly generated optical continuum. These novel fibers allow the improved stabilization of the Cr:forsterite laser to an optical reference cavity. Once stabilized, we have been able to generate low noise microwave signals (instability 2e-15 @10 s) with low residual uncertainty (6e-17). Supplemental Task Progress (submitted 11/2007) Although funding did not arrive until mid way through FY06 (the final year of this project), we continued to make significant progress towards goals outlined in the proposal. Summaries of specific achievements are given below: 1. Low threshold self-reference femtosecond Ti:sapphire laser: The Ti:Sapphire based femtosecond laser frequency comb (FLFC) has shown tremendous utility for a variety of high precision optical frequency measurements. However, because the typical pumping scheme involves a high-power (5-8 W) frequency-doubled Nd:vanadate laser, a complete Ti:Sapphire FLFC remains a costly, heavy, and electrically inefficient device. This is unfortunate because the small footprint of a high repetition rate (500-1000 MHz) Ti:Sapphire laser already lends itself to being the basis of a compact FLFC. These issues have motivated us to develop a more efficient high repetition-rate Ti:Sapphire based FLFC that could be pumped with a smaller 532 nm source. Specifically, we demonstrate a femtosecond ring Ti:Sapphire laser

having frep=585 MHz and a mode-locking threshold requiring as little as 340 mW of 532 nm pump power. At the typical operating conditions, 830 mW of pump power generates 150 pJ pulses, of which ~55 pJ is coupled into a nonlinear microstructure fiber to produce an octave-spanning continuum extending from below 500 nm to above 1000 nm. This continuum is used in a typical f-2f self-referencing scheme to measure fo with S/N greater than 40 dB in 300 kHz bandwidth. This system achieves a beat signal (fbeat) against a single-frequency 1064 nm source with greater than 30 dB S/N in 300 kHz bandwidth. Both f0 and fbeat are phase-locked as required to generate a phase stable frequency comb. Such low threshold operation should lead to smaller, less expensive, and more efficient Ti:Sapphire pumping schemes including either a 1 W Nd-based green sources or an efficient frequency-doubled Yb fiber lasers.

2. Improved octave-spanning Ti:sapphire laser for optical frequency measurements and comparison: To date, octave-spanning lasers (i.e. femtosecond lasers that emit an octave spectrum directly from the oscillator) have been demonstrated with repetition rates up to only 200 MHz. Higher repetition rate lasers, which provide more optical power per mode, are preferable for optical frequency metrology. Additionally, octave-spanning lasers provide little light below 600 nm, which makes measurements in this regime difficult. This is a particular issue at NIST, where three of the four current optical clock standards have local oscillators in this range: Al+ (534 nm), Hg+ (563 nm) and neutral Yb (578 nm). We have developed techniques for simultaneous measurements at these wavelengths with a variable repetition rate (550 MHz – 1.35 GHz) femtosecond laser frequency comb (FLFC) that has an octave-spanning bandwidth required for “f-2f” self-referencing. We also demonstrate that the FLFC can effectively transfer and reproduce the very low frequency noise of the optical standard to which it is stabilized.

3. Continued tests of the residual stability and uncertainty of femtosecond laser frequency combs: We have measured and analyzed the currently known relative frequency uncertainty of femtosecond laser frequency combs (FLFCs) based on Kerr-lens mode-locked Ti:sapphire lasers. Broadband frequency combs generated directly from the laser oscillator, as well as octave-spanning combs generated with nonlinear optical fiber were studied. The relative frequency uncertainty introduced by an optically-referenced FLFC is measured for both its optical and microwave outputs. We find that the relative frequency uncertainty of the optical and microwave outputs of the FLFC can be as low as 8e-20 and 1.7e-18, with a confidence level of 95%, respectively. Photo-detection of the optical pulse train introduces a small amount of excess noise, which degrades the stability and subsequent relative frequency uncertainty limit of the microwave output to 2.6e-17.

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